

CROSSVILLE, TN, USA

WMO: 723250

Lat: 35.951N

Lon: 85.081W

Elev: 569

StdP: 94.67

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03847

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-9.0	-17.5	0.9	-8.3	-14.9	1.1	-7.1	9.0	7.0	8.2	6.5	2.2	320	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	31.1	22.2	29.9	21.9	28.8	21.6	23.9	28.4	23.3	27.7	22.8	27.0	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.4	26.3	22.0	17.9	25.8	21.3	17.2	25.0	74.4	28.5	72.1	27.8	69.8	27.0	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.7	5.7	DB	-16.2	32.7	4.0	2.0	-19.0	34.1	-21.4	35.3	-23.6	36.4	-26.5	37.8	(4)
(5)			WB	-16.7	24.9	3.8	1.0	-19.4	25.6	-21.6	26.2	-23.8	26.8	-26.6	27.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	2.3	4.4	8.5	13.8	18.0	21.8	23.5	23.2	20.0	14.2	8.4	4.4	(6)	
	DBStd	8.77	6.64	5.99	5.68	4.82	3.83	2.50	1.99	2.18	3.27	4.53	5.14	5.75	(7)	
	HDD10.0	834	251	171	96	20	2	0	0	0	15	89	190		(8)	
	HDD18.3	2291	498	390	306	147	53	5	0	1	22	139	298	432	(9)	
	CDD10.0	2143	11	14	50	133	249	354	419	409	299	146	41	16	(10)	
	CDD18.3	559	0	0	1	11	43	109	161	152	71	11	1	0	(11)	
	CDH23.3	3872	0	0	6	71	282	742	1128	1087	487	67	2	0	(12)	
	CDH26.7	944	0	0	0	3	34	172	303	311	113	7	0	0	(13)	
(14) Wind	WSAvg	2.3	2.9	2.9	3.0	2.9	2.2	1.8	1.6	1.5	1.6	2.0	2.4	2.8	(14)	
(15) Precipitation	PrecAvg	1384	118	112	148	121	135	111	130	102	96	77	121	135	(15)	
	PrecMax	1767	256	267	390	286	297	359	304	228	229	183	216	324	(16)	
	PrecMin	978	20	21	36	40	40	22	36	23	17	0	44	24	(17)	
	PrecStd	230	60	47	82	64	59	71	65	50	55	48	47	68	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.9	19.8	24.0	27.2	29.0	31.3	32.5	32.9	31.3	27.3	22.8	19.8	(19)	
		MCWB	14.2	14.7	15.5	16.8	20.4	22.2	22.9	22.2	20.1	19.0	16.1	16.0	(20)	
	2%	DB	16.2	17.7	22.0	25.4	27.6	29.8	30.9	31.1	29.3	25.1	20.4	17.4	(21)	
		MCWB	13.5	13.3	14.1	16.1	19.3	21.9	22.8	22.4	20.3	17.9	14.3	14.2	(22)	
	5%	DB	14.3	15.9	19.6	23.8	26.4	28.7	29.5	29.6	27.9	23.5	18.4	15.3	(23)	
		MCWB	11.7	11.7	13.2	15.5	18.8	21.5	22.6	21.9	20.1	16.7	13.9	12.6	(24)	
	10%	DB	12.3	13.5	17.6	22.1	25.0	27.5	28.4	28.3	26.3	21.8	16.8	13.2	(25)	
		MCWB	10.0	9.8	12.2	14.7	18.2	21.0	22.2	21.8	19.6	16.3	13.1	10.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.7	16.4	16.9	19.3	22.0	24.1	24.9	24.7	22.8	21.1	17.9	16.9	(27)	
		MCDB	17.4	18.6	20.9	24.3	26.6	28.5	29.4	29.5	27.0	24.8	20.5	18.8	(28)	
	2%	WB	14.1	14.8	15.7	18.1	20.8	23.2	24.0	23.7	21.9	19.7	16.6	15.0	(29)	
		MCDB	15.7	17.1	19.7	22.8	25.4	27.8	28.3	28.5	26.3	23.1	19.0	16.9	(30)	
	5%	WB	12.4	12.8	14.6	17.0	20.1	22.6	23.5	23.0	21.3	18.6	15.2	13.1	(31)	
		MCDB	14.0	14.6	18.4	21.3	24.4	26.9	27.8	27.5	25.5	21.7	17.5	14.8	(32)	
	10%	WB	10.4	10.7	13.2	16.0	19.4	21.8	22.9	22.5	20.6	17.5	13.5	11.2	(33)	
		MCDB	11.8	12.8	16.7	20.2	23.4	25.8	27.1	26.7	24.5	20.2	16.0	12.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.5	10.6	11.2	10.8	10.0	9.3	9.9	10.7	11.3	10.5	8.8	(35)	
		MCDBR	9.5	11.2	12.6	12.7	12.0	11.6	10.5	11.5	12.6	12.8	11.8	9.9	(36)	
	5% WB	MCWBR	7.1	7.6	7.1	6.1	5.4	5.1	4.4	4.6	5.2	6.5	6.9	7.3	(37)	
		MCWBR	8.0	9.1	10.3	10.2	9.9	9.8	9.2	9.4	9.7	9.4	8.9	8.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.300	0.312	0.339	0.376	0.421	0.452	0.481	0.490	0.396	0.353	0.317	0.297	(40)	
		taud	2.524	2.480	2.440	2.326	2.225	2.166	2.114	2.074	2.323	2.441	2.506	2.548	(41)	
	Edn at Noon		905	931	931	907	866	835	808	792	858	871	875	886	(42)	
			80	93	106	126	141	150	157	160	118	96	81	74	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.22	2.78	3.94	5.12	5.68	6.14	5.81	5.44	4.61	3.65	2.63	1.92	(44)	
	RadStd		0.19	0.36	0.39	0.36	0.45	0.40	0.33	0.28	0.44	0.47	0.29	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+87	+65	(46)	
(47)	Regional (24 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	+92	(47)	

Nomenclature: See separate page